

# CFB 27 Player XP (Dynasty)

Patch 3 tuning (dynasty-tuning-binary.FTC) + save DYNASTY-2026OFFLINEFINAL.

Players do not have a Level field. Progression is:

earn XP → hit XP threshold → gain Skill Points → spend SP on ratings

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## 1. How is XP calculated?

Base XP comes from offseason progression, games, coach bonuses, and other progression events. Most grant logic is native code (`IncreaseExperiencePoints`) rather than readable `FranTk`.

What *is* readable and confirmed for dynasty:

### Flat / percent add-ons

| Source              | Effect                                                                                      |
|---------------------|---------------------------------------------------------------------------------------------|
| Athletic facilities | +0 to +20 XP by letter grade (A+ = 20)                                                      |
| Coach XP talents    | percent (e.g. Everybody Eats up to +10%) or flat — see docs/coaching-abilities-player-xp.md |

### Difficulty / Wear & Tear XP modifiers — not used for dynasty offseason

`PlayerProgressionTuning` still contains:

| Field                                        | Stored value |
|----------------------------------------------|--------------|
| <code>SkillLevelXPModifierFreshman</code>    | 0.50         |
| <code>SkillLevelXPModifierVarsity</code>     | 0.75         |
| <code>SkillLevelXPModifierAllAmerican</code> | 1.00         |
| <code>SkillLevelXPModifierHeisman</code>     | 1.25         |
| <code>WearAndTearXPModifierOn</code>         | 1.00         |
| <code>WearAndTearXPModifierOff</code>        | 0.40         |

Native code null-checks `skillLevelModRef / wearAndTear*ModRef` near `ProgressPlayers`, so the fields are loaded. But **in-game A/B testing across multiple dynasty players/positions showed no meaningful offseason XP difference** when difficulty or Wear & Tear was changed. Treat these as unused for dynasty training-results XP (likely RTG / leftover shared tuning), not live dynasty multipliers.

### Offseason training results (decoded)

Training-results XP is awarded by native `PlayerProgressionEval::AwardOffSeasonXPToPlayer` (also tied to `IssuePostSeasonTrainingRequest`). Each player gets one roll:

```
base = random(MinXP[year][pos], MaxXP[year][pos])
// redshirt players use RedshirtMinXP / RedshirtMaxXP instead
```

```

xp    = base
      × DevTraitSpline / 100
      × (100 + coachPercent) / 100
      + facilitiesFlat

```

School years: Freshman=0, Sophomore=1, Junior=2, Senior=3.

Tables live in tuning `PositionValueTable` (rows 11–20 normal max/min, 22–31 redshirt max/min), referenced from `save PlayerProgressionEval.MinXPTables / MaxXPTables / redshirt variants`.

### Base XP ranges (non-redshirt, before multipliers)

| Pos       | FR            | SO            | JR            | SR            |
|-----------|---------------|---------------|---------------|---------------|
| QB        | 35,150–52,250 | 31,350–48,450 | 27,550–44,650 | 25,650–40,850 |
| HB        | 37,050–47,500 | 33,250–43,700 | 29,450–39,900 | 25,650–36,100 |
| FB        | 15,600–31,200 | 14,625–29,250 | 12,675–25,350 | 11,700–23,400 |
| WR        | 30,400–51,300 | 29,450–46,550 | 28,500–44,650 | 27,550–42,750 |
| TE        | 25,350–40,950 | 24,375–39,000 | 22,425–35,100 | 21,450–33,150 |
| LT/RT     | 33,150–52,650 | 32,175–50,700 | 31,200–48,750 | 30,225–46,800 |
| LG/C/RG   | 32,175–50,700 | 31,200–48,750 | 30,225–46,800 | 29,250–44,850 |
| LE/RE     | 30,225–44,850 | 29,250–42,900 | 27,300–39,000 | 26,325–37,050 |
| DT        | 29,250–44,850 | 28,275–42,900 | 27,300–40,950 | 25,350–37,050 |
| LOLB/ROLB | 26,325–41,925 | 24,375–39,975 | 22,425–36,075 | 20,475–32,175 |
| MLB       | 21,450–35,100 | 19,500–33,150 | 17,550–29,250 | 16,575–25,350 |
| CB        | 31,350–49,400 | 30,400–47,500 | 28,500–45,600 | 28,500–43,700 |
| FS        | 29,450–49,400 | 28,500–47,500 | 26,600–43,700 | 25,650–41,800 |
| SS        | 29,450–45,600 | 28,500–43,700 | 26,600–39,900 | 25,650–38,000 |
| K/P       | 4,750–9,500   | 4,500–9,000   | 4,250–8,500   | 4,000–8,000   |
| LS        | 0             | 0             | 0             | 0             |

Redshirt Min/Max bands are about **5–8% lower** than the matching non-redshirt band.

Relevant coach % talents on this grant (stack additively; see `docs/coaching-abilities-player-xp.md`):

| Talent                  | Typical max stack |
|-------------------------|-------------------|
| Put in Work (offseason) | +10%              |
| Everybody Eats (all XP) | +10%              |
| Star Maker (starters)   | +17%              |
| Whisperer (FR/SO)       | +17%              |

### Worked examples (no coach talents)

| Player                 | Approx mid base | After trait     |
|------------------------|-----------------|-----------------|
| FR QB, Normal          | ~43,700         | ~ <b>43,700</b> |
| FR QB, Elite           | ~43,700         | ~ <b>65,550</b> |
| FR QB, Elite, max roll | 52,250          | ~ <b>78,375</b> |
| FR K, Normal           | ~7,125          | ~ <b>7,125</b>  |

## 2. How do Dev Traits affect XP gain?

The live CFB offseason progression path uses `PlayerDevTraitsSpline`, divided by `DevTraitDivisor = 100`:

| Dev trait  | XP multiplier |
|------------|---------------|
| Normal (0) | <b>1.00x</b>  |
| Impact (1) | <b>1.20x</b>  |
| Star (2)   | <b>1.35x</b>  |
| Elite (3)  | <b>1.50x</b>  |

Ghidra confirms `AwardOffSeasonXPToPlayer` reads a valid `playerDevTraitInt`, and startup validation requires `mDevTraitSpline` to be attached. The matching trait-indexed spline in the CFB data is:

X = [0, 1, 2, 3]  
Y = [100, 120, 135, 150]

This confirms the modifiers for **offseason progression XP**. It does not prove that every other XP source applies the same trait multiplier.

### Do not use the Madden weekly-goal values

`PlayerWeeklyGoal.DevelopmentSpline` contains 0.75x / 1.00x / 2.00x / 5.00x plus an `AgeSpline`. CFB dynasty's `PlayerGoalTable` is empty for every position, so that weekly-goal path is not the live dynasty progression formula.

## 3. How much XP to earn a level?

Use `PlayerSPThresholdSpline` (OVR → XP needed). This is the only populated XP-cost curve for dynasty players. (`PlayerXPLevelSpline` is all zeros / unused.)

| Player OVR | XP to level |
|------------|-------------|
| 50–60      | <b>1500</b> |
| 65–70      | <b>1750</b> |
| 75–80      | <b>2000</b> |
| 85–90      | <b>2250</b> |
| 95–99      | <b>2500</b> |

Higher-OVR players need more XP per step.

## 4. How many Skill Points per level-up?

**1 Skill Point per level-up** in dynasty.

Evidence: `PlayerProgressionTuning.IncrementAmount = 1`.

(RTG is different: `RTGPlayerSPPerLevel = 40` — ignore that for dynasty.)

You spend those SP on attribute/ability upgrades (`AbilityProgressionTunable.UpgradeCostSpline`). Cost scales with current rating (often 5 SP at low ratings, rising to 30–45+ near 90–95). Cap note: `AbilityUpgradeCapPerSkillPoint = 5`.

### Quick examples

- **FR QB, Impact, mid roll (~43,700):**  $\times 1.20 \rightarrow \sim 52,440$  XP. At 60 OVR (1500 XP/level) that is ~35 SP before coach/facilities add-ons.
- **Difficulty / Wear & Tear:** changing these does **not** move dynasty offseason training XP in A/B tests.
- **90 OVR:** still needs **2250 XP** per level  $\rightarrow$  **1 SP** each time the threshold is hit.

## Sources

| Question                | Primary field / table                                                                                   |
|-------------------------|---------------------------------------------------------------------------------------------------------|
| Offseason base XP       | <code>PlayerProgressionEval</code> Min/Max/Redshirt XP tables →<br><code>PositionValueTable</code>      |
| Coach XP add-ons        | coach <code>Talent</code> / <code>TalentDataInt</code>                                                  |
| Difficulty / W&T fields | present on <code>PlayerProgressionTuning</code> , <b>not live for dynasty offseason</b><br>(A/B tested) |
| XP to level             | <code>PlayerSPThresholdSpline</code>                                                                    |
| SP per level            | <code>IncrementAmount</code> (= 1)                                                                      |
| DevTrait offseason XP   | Native <code>AwardOffSeasonXPToPlayer</code> + <code>PlayerDevTraitsSpline</code>                       |